

Creating Rules and Monitors

Monitors are used for heaving a health state and also generates alert, based on the state. Rules ae used to alert in SCOM each time the specific event is logged. In this case you have no particular health state, associated with an event, you only alert, based on the event.

With the monitor you can either do a manual rest of the monitor (clear the health state) or you can chose the automated reset.

In System Center - Operations Manager, if a management pack's default settings contain a monitor or rule that is not necessary in your environment, you can use overrides to disable this monitor or rule. In addition, some management packs ship with some rules or monitors disabled; you should read the management pack guide to identify the workflows that are disabled by default and determine if you should enable any of them for your monitoring needs. For example, the management packs for network monitoring contain rules and monitors that are vendor-specific. Many vendor-specific rules and monitors in the network management pack are disabled to avoid performance impact. You should identify the devices used in your environment and use overrides to enable the rules and monitors specific to your devices

- *To enable or disable a monitor or rule using overrides*

1. Log on to the computer with an account that is a member of the Operations Manager Advanced Operator role.
2. In the Operations console, click **Authoring**.
3. In the **Authoring** workspace, click **Monitors** (or **Rules** if you want to disable a rule).
4. In the **Monitors** or **Rules** section, click the monitor or rule that you want to disable.
5. On the Operations console toolbar, click **Overrides** and then point to **Override the Monitor** (or **Rule**). You can choose to override this monitor or rule for objects of a specific type or for all objects within a group. After you choose which group of object type to override, the **Override Properties** dialog box opens, enabling you to view the default settings contained in this monitor or rule.
6. In the **Override Properties** dialog box, click to select the **Override** check box that corresponds to the **Enabled** parameter.
7. In the **Override Setting** column, click **True** to enable the rule or monitor or **False** to disable the rule or monitor.
8. In the **Select destination management pack** list, click the appropriate management pack in which to store the override or create a new unsealed management pack by clicking **New**.
9. When you complete your changes, click **OK**.

- ***To override a monitor or Rule***

1. Log on to the computer with an account that is a member of the Operations Manager Advanced Operator role.
2. In the Operations console, click **Authoring**.
3. In the **Authoring** workspace, expand **Management Pack Objects** and then click **Monitors (or Rules if you want to override rules)**.
4. In the **Monitors** pane, expand an object type completely and then click a monitor.
5. On the Operations console toolbar, click **Overrides** and then point to **Override the Monitor**. You can choose to override this monitor for objects of a specific type or for all objects within a group. After you choose which group of object type to override, the **Override Properties** dialog box opens, enabling you to view the default settings contained in this monitor. You can then choose whether to override each individual setting contained in the monitor.
6. Click to place a check mark in the **Override** column next to each parameter that you want to override. The **Override Value** can now be edited. Change the value in **Override Value** to the value you want the parameter to use.
7. Either select a management pack from the **Select destination management pack** list or create a new unsealed management pack by clicking **New**.
8. When you complete your changes, click **OK**.

1. Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-enable-or-disable-rule-monitor?view=sc-om-2019>
2. Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-mp-override-rule-monitor?view=sc-om-2019>

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Agentless Monitoring

In System Center Operations Manager there are many different ways to install agents to client machines. These methods include:

- Discovery Wizard
- Manual Agent Installation of the SCOM CD
- msixec.exe
- Group Policy
- Operations Manager Command Shell
- Scheduled Discoveries
- AD Integration Auto Assignment
- SCCM Deployment

- **Discovery Wizard**

The discovery wizard is one of the most popular ways for Administrators to deploy agents to new managed assets in a SCOM infrastructure. The discovery wizard broadcasts subnets to find out which devices require agents installed, and gives you the option which devices you want to push the SCOM agent out to over RPC.

There are two types of discoveries:

- Automatic Computer Discovery
- Advanced Discovery

Automatic computer discovery does every PC in the domain.

Advanced discovery lets you select things like just servers. This method is recommended.

- **Manual Agent Installation of the SCOM CD**

You can insert the SCOM installation disk into a windows client and install the Agent simply using the disk. You will then need to approve it in the operations manager console as a managed client. You can use the msixec.exe tool to install SCOM agents on workstations. Using admin tools such as psexec we can run the tool on multiple computers at a given time for mass deployments. By default all these computers will appear under pending management. If you want them to automatically be approved there is an option under Operations Manager in global settings.

- **Group Policy**

You can push the agent out to PC's using standard software deployment through group policy. When using group policy machines will not automatically be approved and you will have to add them manually unless you Auto-approve new manually installed agents.

- **Operations Manager Command Shell**

You can install the SCOM agent on a single workstation using the Ops Mgr Shell by using a command similar to:

```
install-agent -ManagementServer servername -AgentManagedComputer agentcomputername
```

Or you could make a very simple powershell (.ps1) script. Which will mass deploy agents out to multiple machines.

- **Scheduled Discoveries**

Scheduled discoveries was the preferred way of deploying agents but this feature was lost moving to SCOM 2007 and was more or less replaced by AD Integration. However we can create a powershell script that we can schedule to bring this feature back if you wish to do your deployments this way.

- **AD Integration Auto Assignment**

This is the most automated way to do the installation of agents on your network. If you wish to use AD Integration you need to prepare your domain using MOMADAdmin.exe which can be found on the SCOM CD in the support folder. This commandline tool creates a OperationsManager container on your domain partition. Auto Assignment is done using LDAP queries. You can simply use something like AD Users and Computers which has a nice interface to create LDAP queries for you

- **SCCM Deployment**

You can also deploy scom agents using management software such as Microsoft System Center Configuration Manager server. You need to create package in MSI or EXE form, once your package is ready you can deploy through SCCM on All or Selected systems...

Ref: <http://clintboessen.blogspot.com/2009/05/ways-to-deploy-agents-in-scom.html>

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Creating Groups

Groups are logical collections of objects, such as Windows-based computers, hard disks, or instances of Microsoft SQL Server. You create a group by using the Create Group Wizard. You can explicitly assign membership to a group or you can create rules that will generate a dynamic group membership.

There are two types of groups:

Computer groups only contain computers. Instance groups can contain all object types, such as an instance of a health service or an instance of a SQL database. Both computer groups and instance groups can contain other computer and instance groups. Another way to view the difference between the group types is:

1. An instance group is populated with objects that match your criteria.
2. A computer group is populated by computers that host objects that match your criteria.

Some of the purposes for using groups are:

- To scope overrides to a specific subset of computers.
- To scope alert notifications or product connector subscriptions for a specific set of computers.
- To scope user consoles, so the user role only sees the servers they are responsible for.
- To scope a set of computers that need to go into a scheduled maintenance mode.
- To scope application views only to computers that host a given application.
- To create a rollup health state view of an otherwise unrelated set of computers.
- To create a set of computers for a report.

➤ **To create a group in Operations Manager**

1. Log on to the computer with an account that is a member of the Operations Manager Administrators role.
2. In the Operations console, click **Authoring**.
3. Right-click **Groups**, and then click **Create a new Group** to start the Create Group Wizard.
4. On the **Enter a name and description for the new group** page, do the following:
 - a) Type the **Name** for the group.
 - b) Optionally, type the **Description** for the group. A description of the group membership makes it easier to select the right group for views, overrides, and so forth.
 - c) Select a **destination management pack** from the list, or click **New** to create a management pack with the Create a Management Pack Wizard.
 - d) Click **Next**.
5. On the **Explicit Members - Choose Members from a List** page, you can add explicit objects to the group.

6. On the **Dynamic Members - Create a Membership Formula** page, you can add a dynamic membership formula to the group or click **Next** to continue to the **Subgroups** page.
7. On the **Choose Optional Subgroups** page, either click **Next** to not add groups to the group, or click **Add/Remove Subgroups** to add groups.
8. On the **Excluded Members - Specify Exclude List** page, click **Finish** to not exclude objects from the group.

Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-create-manage-groups?view=sc-om-2019>

Configuring Notifications

An alert can actually provide us information a failure in a software or in the hardware, but it may also be a configuration error that we may have missed when setting up something.

Alerts help us to identify ongoing or upcoming issues that we may get within our data center, it may be either software or hardware-related issues. It is important to fully understand the meaning of the alert(s) that we receive before taking any action.

- Alert Classification

All alerts in SCOM have a classification, this means that each alert has its own priority, it will also have its own severity depending on how serious the issue is.

- Alert Severity

There are a total of three (3) different alerts in SCOM and each severity has its own value, these numeric values are more commonly seen when creating overrides for either monitors or rules.

- Alert Priority

Similar to the alert severities, there are also a total of three (3) different alert priorities in SCOM:

Alert priority is important because it helps us determine how an alert should be processed, all management packs have a configured alert priority for the alerts that may rise, but these priorities may not always be relevant according to our own environment and needs, therefore it's important that each organization checks the priorities and configures them based on their environment and needs.

- **Alert Generation**

In SCOM there are two (2) possible ways of generating alerts:

1. Alert generated by a **Monitor**
2. Alert generated by a **Rule**

SCOM uses conditions for generating alerts, when a specific condition or multiple conditions are met, only then will an alert be raised.

The condition could be for example:

- Whenever a specific event in the Windows event log occurs, we raise an alert.
- Whenever a service or operation is failing, we raise an alert.

➤ **Alert Notifications using Channels**

When SCOM generates an alert, the alert will be shown in an *Alert View* or in a *Dashboard* under the **Monitoring** pane in the Operations Console, SCOM can also notify individuals by sending the alert as a notification.

SCOM uses "channels" this is the so-called engine that will decide what type of notification we want to send, there are four (4) different ways of sending notifications in SCOM:

1. Email (SMTP) Notification Channel

The email notification is the most commonly used way of sending notifications, SCOM supports sending emails by using both internal and external email authentications.

Since SCOM 2019 there has been a lot of enhancements in the email notifications, the subscription criteria has also been enhanced, SCOM 2019 now also supports email notifications in HTML format.

2. Instant Message (IM) Notification Channel

Notifications can also be sent as instant messages, many organisations are using Skype for Business (previously known as Lync) as the main instant messaging communications program, with SCOM we can achieve to send instant messages directly to users on Skype for Business.

3. Text Message (SMS) Notification Channel

SCOM is also able to send text messages, more commonly known as "SMS" (Short Message Service). For this to work, we'll need either an SMS module or a modem into which we'll need to insert a SIM card to be able to send SMS to recipients.

4. Command Notification Channel

SCOM is also able to make use of the commands, this can be used to run executable programs or scripts in response to an alert.

We can for example write to an event log, create a text file with the alert information or create a ticket to a ticketing system, or even use a script to send emails or SMS messages.

➤ **To enable an email notification channel**

1. Log on to the computer with a user account that is a member of the Operations Manager Administrators role.
2. In the Operations console, click **Administration**.
3. In the navigation pane, under **Notifications**, right-click **Channels**. Click **New channel** and then click **E-mail (SMTP)**.
4. Type a name for the channel, such as **SMTP channel** and optionally provide a description. Click **Next**.
5. In the **SMTP servers** area, click **Add**.
6. In the **Add SMTP Server** dialog box, type the fully qualified domain name (FQDN) of a Simple Mail Transfer Protocol (SMTP) server, type the port number, select the authentication method used by the SMTP server, and then click **OK**.
7. Type the **Return Address** that should appear on email notifications, and then in the **Retry interval** list, select the number of minutes to wait before trying to resend a notification to the primary SMTP server. Click **Next**.
8. In the **Default e-mail notification format** area, specify the **E-mail subject** and **E-mail message** text or leave the default selections, select the **Importance** level that you want the emails sent with, and then specify the **Encoding** type. You can click the right arrow next to the **E-mail subject** and **E-mail message** boxes for a full list of available variables.
9. Click **Finish**, and then click **Close**.

1. Ref: <https://www.cookdown.com/blog/scom-alert-basics>

2. Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-notifications-create-email-channel?view=sc-om-2019>

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Installing SQL Reporting Services

SQL Server Reporting Services (SSRS) provides a set of on-premises tools and services that create, deploy, and manage mobile and paginated reports. SQL Server Reporting Services installation involves server components for storing report items, rendering reports, and processing of subscription and other report services.

➤ **Installing report server**

Installing a report server is straightforward. There are only a few steps to install the files.

1. Find the location of SQLServerReportingServices.exe and launch the installer.
2. Select **Install Reporting Services**.
3. Choose an edition to install and then select **Next**.
4. Read and agree to the license terms and conditions and then select **Next**.
5. You need to have a Database Engine available to store the report server database. Select **Next** to install the report server only.
6. Specify the install location for the report server. Select **Install** to continue.
7. After a successful setup, select **Configure Report Server** to launch the Report Server Configuration Manager

➤ **Configure your report server**

After you select **Configure Report Server** in the setup, you will be presented with **Report Server Configuration Manager**. You need to [create a report server database](#) to complete the initial configuration of Reporting Services. A SQL Server Database server is required to complete this step.

➤ **Creating a database on a different server**

If you are creating the report server database on a database server on a different machine, you need to change the service account for the report server to a credential that is recognized on the database server.

By default, the report server uses the virtual service account. If you try to create a database on a different server, you may receive the following error on the Applying connection rights step.

1. Ref: <https://docs.microsoft.com/en-us/sql/reporting-services/create-deploy-and-manage-mobile-and-paginated-reports?view=sql-server-ver15>
2. Ref: <https://docs.microsoft.com/en-us/sql/reporting-services/install-windows/install-reporting-services?view=sql-server-ver15>

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Installing Operations Manager Reporting

In this procedure, the Reporting server is installed on a stand-alone server that is hosting the SQL Server database and SQL Server Reporting Services.

➤ Installing Operations Manager reporting

SQL Server Reporting Services installed on this instance of SQL Server can be used only by Operations Manager. If any reports exist on this SSRS instance, the System Center Operations Manager Reporting Services installer overrides all of the data/reports on it. Ensure that SQL Server Reporting Services has been correctly installed and configured.

1. Log on to the computer with an account that has local administrative credentials.
2. On the Operations Manager installation media, run **Setup.exe**, and then click **Install**.
3. On the **Getting Started, Select features to install** page, select the **Reporting server** feature. To read more about each feature and its requirements, click **Expand all**, or expand the buttons next to each feature, and then click **Next**.
4. On the **Getting Started, Select installation location** page, accept the default value, or type a new location or browse to one, and then click **Next**.
5. On the **Prerequisites** page, review and resolve any warnings or errors, and then click **Verify Prerequisites Again** to recheck the system.
6. If the Prerequisites checker does not return any warnings or errors, continue to the **Prerequisites, Proceed with Setup** page. Click **Next**.
7. On the **Configuration, Specify a Management server** page, enter the name of a management server that is used by the Reporting features only. Then click **Next**.
8. On the **Configuration, SQL Server instance for reporting services** page, select the instance of SQL Server that hosts SQL Server Reporting Services, and then click **Next**.
9. On the **Configuration, Configure Operations Manager accounts** page, enter the credentials for the **Data Reader account**, and then click **Next**.
10. On the **Configuration, Help improve System Center - Operations Manager** page, select your options, and then click **Next**.
11. If Windows Update is not activated on the computer, the **Configuration, Microsoft Update** page appears. Select your options, and then click **Next**.
12. Review the options on the **Configuration, Installation Summary** page, and then click **Install**. Setup continues.
13. When Setup is finished, the **Setup is complete** page appears. Click **Close**.

➤ ***To confirm the health of Operations Manager reports***

1. Open the Operations console, and select the **Reporting** workspace.
2. Click **Microsoft ODR Report Library**, and then double-click any of the reports listed. The selected report is then generated and displayed in a new window.

By default, you should see the following reports:

- Alerts Per Day
 - Instance Space
 - Management Group
 - Management Packs
 - Most Common Alerts
3. Close the report window.

Ref: <https://docs.microsoft.com/en-us/system-center/scom/deploy-install-reporting-server?view=sc-om-2019>

Creating Reports

Note: When asked along with viewing and customizing reports, write any one procedure for creating report. If creating reports is asked for 10 marks, write minimum 2 procedures for creating reports.

Reports helps us to review the operational telemetry and configuration information to help troubleshoot issues or problems, review the day-to-day health of your IT services, and make decisions that drive changes in capacity planning or service operations of the service. There are several types of reports in SCOM 2012:

- Operational data report
- Event analysis report
- Availability report
- Alerts report
- Alert logging latency report

➤ ***Operational data report:***

The Microsoft Customer Experience Improvement Program (CEIP) collects information about how you use Microsoft programs and about some of the issues you might encounter. Microsoft uses this information to improve the products and features you use most often and to help solve issues. Participation in the program is strictly voluntary.

During setup of Operations Manager Reporting, on the **Operational Data Reports** page, you had the option to join CEIP. If you elected to join CEIP, Operations Manager Reporting collects information about your installation and sends reports to Microsoft on a weekly basis. You can view the contents of these Operational Data Reports by creating a Microsoft ODR Report.

Create an Operational Data report

1. Log on to the computer with an account that is a member of the Operations Manager Report Operators role.
2. In the Operations console, click **Reporting**.
3. In the **Reporting** workspace, expand **Reporting**, and then click **Microsoft ODR Report Library**.
4. In the **Reports** pane, right-click one of the reports (for example, **Management Packs**), and then click **Open**.
5. In the **Report** view, in the **Parameter** area, click the down arrow in the **From** box, point to **This week**, and then click **Sunday**.
6. Click the down arrow in the **To** box, point to **This week**, and then click **Saturday**.
7. Click **Run** to display the ODR Report.
8. Click **Close** to close the report.

➤ **Availability report**

The following procedure is an example of how you create an availability report for a managed computer. The procedure presented here is applicable to creating other types of availability reports. In this example procedure, you generate a report for the entire week.

The availability report provides the following information about the selected computers:

- **Down** - computer state is critical (red)
- **Up** - computer state is healthy (green)
- **Yellow** - computer state is warning (yellow)
- **Unmonitored** - computer or monitor did not exist during reporting period
- **Monitor disabled** - monitor has been disabled, such as by using an override
- **Monitoring unavailable** - the System Center Management Health service monitoring the computer is unavailable
- **Planned/unplanned maintenance** - computer is in maintenance mode; overrides all other states

Create an Availability report

1. Log on to the computer with an account that is a member of the Operations Manager Administrators role.
2. In the Operations console, click **Monitoring**.
3. In the **Monitoring** workspace, expand **Monitoring**, and then click **Windows Computers**.
4. In the **Windows Computers** pane, click the row, or rows, that represent the computer for which you want to run an availability report.
5. In the **Tasks** pane, under **Report Tasks**, click **Availability**.
6. In the **Report** view, in the **Parameter** area, click the down arrow in the **From** box, point to **This week**, and then click **Sunday**.
7. Click the down arrow in the **To** box, point to **This week** and then click **Saturday**.
8. Click **Use business hours**.
9. When you have specified the timeframe for the report, click **Run** to display the Availability Report.

10. For a more detailed report, such as a report showing a graph for every day, click the horizontal bar graph under **Availability Tracker**.
11. In the tool bar, click **View**, point to **Go To**, and then click **Back to Parent Report** to return to the original report.
12. Click **Close** to close the report.

➤ **Alert Report:**

An alerts report summarizes alerts that have occurred on a managed entity. The following procedure is an example of how you create an alerts report for a managed computer. The procedure presented here is applicable to creating other types of alerts reports. In this example procedure, you generate a report for the previous 24-hour period.

Create an Alerts report

1. Log on to the computer with an account that is a member of the Operations Manager Administrators role.
2. In the Operations console, click **Monitoring**.
3. In the **Monitoring** workspace, expand **Monitoring**, and then click **Windows Computers**.
4. In the **Windows Computers** pane, click a row with a Health Service instance.
5. In the **Tasks** pane, under **Report Tasks**, click **Alerts**.
6. In the **Reporting Parameter** area, click the down arrow in the **From** box and then click **Yesterday**. Click **Run** to display the **Alert Report**.
7. Click **Close** to close the report.

Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-reports-create-reports?view=sc-om-2019>

Customizing Reports

Custom reports are intended for advanced users who are comfortable creating their own reports by using Microsoft SQL Server Reporting Services and SQL Server Business Intelligence Development Studio, SQL Server Reporting Services Report Builder, or Microsoft Visual Studio Report Designer.

If report types using prebuilt queries do not deliver the reporting data that you need, use *custom reports* by creating the queries yourself to retrieve just the information you want. For example, you can create a query that consolidates the data from multiple objects into a single report. You can then customize the appearance of the report, using one of several tools or programs that include Microsoft SQL Server 2008 Business Intelligence Development Studio, Visual Studio Report Designer, Microsoft SQL Server Reporting Services Report Builder, or Excel,

Before you create a custom report, you must first decide what information you need in the report. Determine what counters you want to monitor, and what parameters you want to report on. For example, if you are reporting on availability, what time period should the report cover? If you are

reporting on performance, what is the acceptable threshold for the counter? Next, determine the appearance of the report.

To write a custom report, you must install the following services and programs on your report development server:

- System Center Operations Manager 2007 Reporting Services
- SQL Server 2005 Reporting Services or SQL Server 2008 Reporting Services
- SQL Server 2005 Business Intelligence Development Studio or SQL Server 2008 Business Intelligence Development Studio, SQL Server 2005 Reporting Services Report Builder or SQL Server 2008 Reporting Services Report Builder, or Visual Studio 2008 Report Designer

To write a custom report, follows these basic steps:

1. Write a Transact-SQL query that retrieves the information that you need.
2. Add parameters to the query for operators to use to run the report.
3. Design the report in SQL Server Report Designer or Visual Studio Report Designer.
4. Optionally, load the report definition and Report Definitions Language (RPDL) file into a management pack for distribution.
5. Import the report or the management pack.

1. Ref: [https://docs.microsoft.com/en-us/previous-versions/system-center/operations-manager-2007-r2/gg508710\(v=technet.10\)?redirectedfrom=MSDN](https://docs.microsoft.com/en-us/previous-versions/system-center/operations-manager-2007-r2/gg508710(v=technet.10)?redirectedfrom=MSDN)
2. Ref: [https://docs.microsoft.com/en-us/previous-versions/system-center/operations-manager-2007-r2/gg542269\(v=technet.10\)](https://docs.microsoft.com/en-us/previous-versions/system-center/operations-manager-2007-r2/gg542269(v=technet.10))

Viewing Reports

You can view reports in Operations Manager by selecting it, entering the required parameters and executing it. Once you run it, you can also choose to save the configuration to a linked report, which deploys that report with different settings. You can export a report to another file format, such as PowerPoint, Image, PDF, Microsoft Word, or Microsoft Excel for your reporting users.

➤ To run a report from the Reporting pane

Use the following procedure to run a report from the Reporting workspace. In this example, you will run an availability report.

1. Log on to the computer with an account that is a member of the Operations Manager Report Operators role.
2. In the Operations console, click **Reporting**.
3. In the **Reporting** workspace, expand **Reporting** and then click **Microsoft Generic Report Library**.
4. In the **Reports** pane, right-click one of the reports (for example, **Availability**) and then click **Open**.

5. In the **Report** view, in the **Parameter** area, click the down arrow in the **From** box, point to **This week**, and then click **Sunday**.
6. Click the down arrow in the **To** box, point to **This week** and then click **Saturday**.
7. Click **Use business hours**.
8. Click **Add Object**.
9. In the **AddObject** dialog box, in the **ObjectName** text box, type the computer name for a computer that you want to report availability, and then click **Search**.
10. In the **Available items** list, click the computer you want to run a report for, click **Add**, and then click **OK**.
11. Click **Run** to display the Availability Report.

➤ **Saving Report**

The following steps outline the procedure for saving a report to a management pack:

1. Run a report from the Generic Report Library or from another management pack, such as the SQL Server management pack. Specify the parameters you want to for the report and click **Run**.
2. When the report renders, validate that it contains the information you need.
3. On the **File** menu, click **Save to management pack**.
4. Follow the instructions in the wizard to save the report.

➤ **Exporting a Report**

1. After a report has been run, in the toolbar, click the **File** menu, point to **Export**, and then click the format you want to export the file to.
2. In the **Save As** dialog box, select the folder where you want to save the report, and then click **Save**.

Ref: <https://docs.microsoft.com/en-us/system-center/scom/manage-reports-run-save-export?view=sc-om-2016>

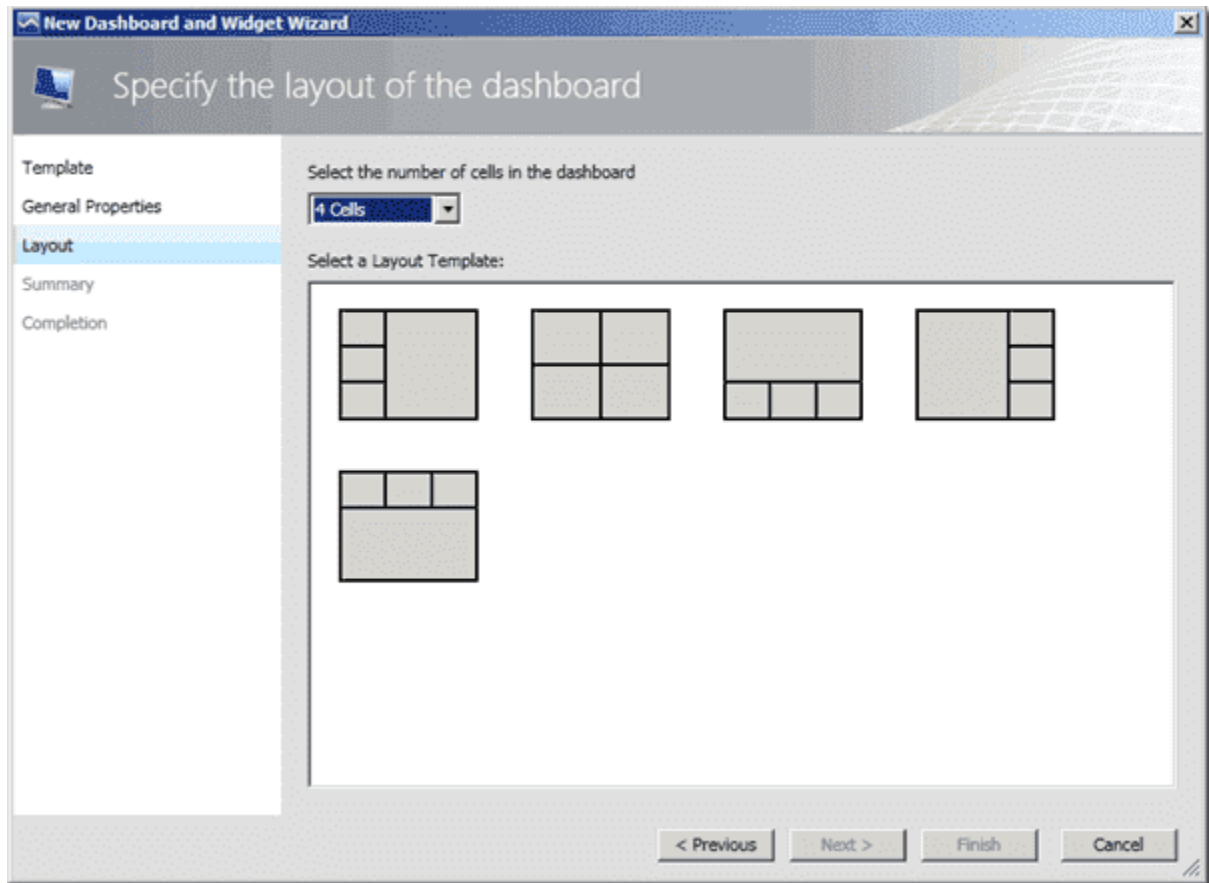
Dashboards

SCOM collects vast amounts of data, we must take care of tasks like filtering and displaying the right data to the right people at the right time.

There are three primary ways of doing this:

- **Alerts** that tell you that something is wrong and needs attention,
- **reports** showing historical data and
- **dashboards** that show actionable, real time data in a visual fashion that can be personalised.

SCOM 2012 supports dashboard wizard for creating dashboards. The wizard is available in both the native console and the web console and the resulting dashboards can be displayed in the Console and the Web Console. SCOM 2012 can have nested dashboards where drilling down into particular data lead to another dashboard.

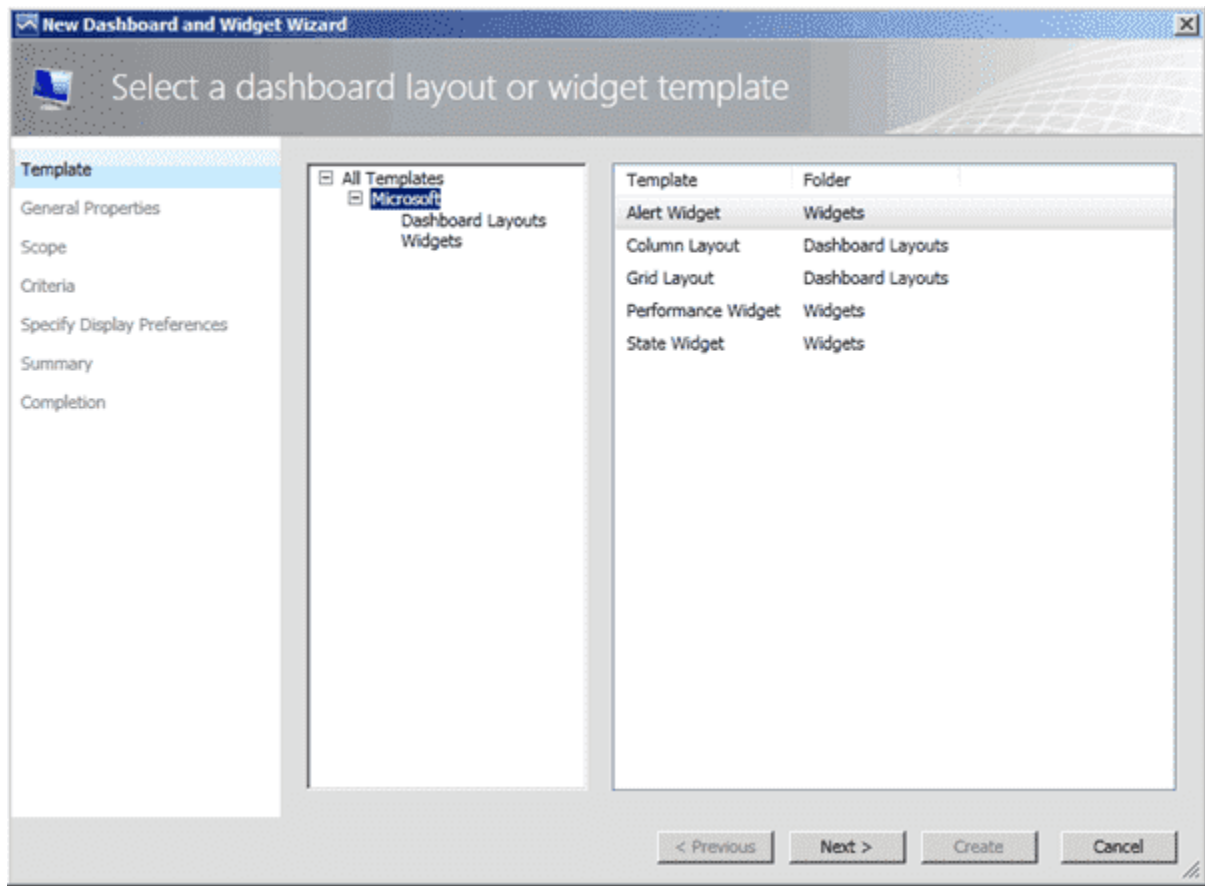


Creating custom Dashboards is not only useful, it's also very easy with the new wizard.

There are three steps to creating a dashboard in SCOM 2012:

- first select a layout based on the number of cells or columns desired;
- add a widget in each cell (types include Alert, Performance and State)
- finally configure each widget with a particular scope and criteria as well as display preferences. T

The Performance widget can now display data from either the Operational or DataWarehouse databases. Apart from the comprehensive built in dashboards third party management packs can add feature packs to support their own widgets. Both SQL Server and Hyper-V have dashboards in the works.



For each cell define what widget you want and configure it's properties.

Personalisation of dashboards by a user are now stored in the database and thus roam with the user to different PCs and environments. Dashboards in the web console all have a distinct URL, this also makes it easy to disseminate information to non-technical users, as they can simply bookmark particular dashboards.

Ref: <https://4sysops.com/archives/scom-2012-review-part-8-dashboards/>

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Considerations for High Availability and Disaster Recovery

System Center – Operations Manager servers and features can potentially fail, impacting Operations Manager functionality. The amount of data and functionality lost during a failure is different in each failure scenario. It depends on the role of the failing feature, the length of time it takes to recover the failing feature.

High availability

High-availability needs are addressed by building redundancy into the management group for the Operations Manager operational and data warehouse databases, etc.

- The multiple servers, can make use of SQL Server Always On for providing high availability and service continuity of the Operations Manager databases.
- Management server fault-tolerance is provided by having at least two management servers and by using the resource pools for monitoring UNIX servers, Linux servers, and network devices.
- Agent-based Windows servers can be configured with a primary and secondary management server to redirect agent communications should a management server fail.
- Operations console connections can be made highly available by configuring high availability for the Data Access Services. This can be done by installing Microsoft Network Load Balancing (NLB) or using a hardware-based load balancers.
- Multiple gateway servers can be deployed across a trust boundary to provide redundant pathways for agents that lie across that trust boundary.

Disaster recovery

Disaster recovery relates to measures taken to ensure that operations can be resumed if a catastrophic failure (for example, loss of the entire data center that hosts the primary infrastructure). Operations Manager will be able to continue supporting proactive monitoring and reporting of the performance and availability of your critical IT services.

The two most common disaster recovery design configurations for Operations Manager are:

- Creating a duplicate management group deployed to your secondary data center that duplicates in scale and configuration, the primary management group.
 - Deploying additional servers in a secondary data center to support the Operational and Data Warehouse database, with management servers deployed in a cold-standby configuration, not participating in the management group until recovery actions need to be performed.
-
- Deploying a duplicate management group is an option when there is no tolerance for downtime; however, it is the most complex option.
 - Integration with other monitoring platforms or ITSM platforms such as System Center - Service Manager, Remedy or ServiceNow will need to exist.
 - Operations Manager can be deployed on Azure virtual machines as an alternative disaster recovery option to maintain continuity of the management group.

- It will be necessary to also deploy SQL Server on a virtual machine in Azure and not in a hybrid configuration, as the inactivity between a management server and the SQL Server hosting the Operations Manager databases will negatively impact performance of the management group.

Ref: <https://docs.microsoft.com/en-us/system-center/scom/plan-hadr-design?view=sc-om-2019>

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Creating Custom Management Packs

Management pack templates

Management Pack Templates provide Monitoring wizards that let you create complete monitoring scenarios with minimal input. The wizard creates the required monitors, rules, and even targets to implement the particular scenario. There's no requirement for you to understand the management pack elements that are created. You can modify the configuration of the wizard itself if you want to change the way that monitoring is being performed.

Conceptual view of a monitoring wizard



If a management pack template is available for your particular monitoring requirement, using the template most likely is your best strategy. In many cases, you could create the individual rules and monitors yourself, but this exercise is more complex than using an available template. In addition, some templates perform actions that you can't perform in any other way in the Operations console.

The following table lists the management pack templates that are part of the standard Operations Manager installation. You can install other management packs that might provide additional templates. Each of these templates is covered in detail in the subsequent sections of this article.

Table 1: Management Pack Templates

Template	Description
.NET Application Performance Monitoring Template	Monitor .NET applications to get details about application performance and reliability.
OLE DB Data Source Template	Monitor a database accessible with OLE DB.
Process Monitoring Template	Discover and monitor instances of a particular Windows process.
TCP Port Template	Monitor the availability of an application that is listening on a specific port.
UNIX or Linux Log File	Monitor a UNIX or Linux log file for a specific entry.
UNIX or Linux Process	Monitor a UNIX or Linux process.
Web Application Availability Monitoring Template	Monitor the availability of one or more web application URLs and run these monitoring tests from internal locations.
Web Application Transaction Monitoring Template	Monitor the availability, operation, and performance of a web application.
Windows Service Template	Discover and monitor instances of a particular Windows service.

Create management pack templates

To create a management pack template, follow these steps:

1. Start the Operations console with an account that has Author credentials in the management group.
2. Open the Authoring workspace.
3. In the Authoring navigation pane, right-click the Management Pack Templates node, and then select Add Monitoring Wizard.
4. Select the management pack template that you want to create, and then select Next.
5. Follow the instructions for the template that you selected. You can use the links in the table above to access this content.

Edit an existing management pack template

To edit an existing management pack template, follow these steps:

1. Start the Operations console with an account that has Author credentials in the management group.
2. Open the Authoring workspace.
3. In the Authoring navigation pane, expand the Management Pack Templates node.
4. Right-click the template to edit and then select Properties.

View the elements created by the management pack template

To view the elements created by the management pack template, follow these steps:

1. Start the Operations console with an account that has Author credentials in the management group.
2. Open the Authoring workspace.
3. In the Authoring navigation pane, expand the Management Pack Templates node.
4. Right-click the template to view, select View Management Pack Objects , and then select the kind of element that you want to view.

Custom Alerts allow you to generate alerts in Operations Manager when a specific condition or event occurs in your systems, applications, or logs. These alerts notify administrators so they can quickly respond to issues.

1. What a Custom Alert Is

A custom alert is an alert that you define yourself, instead of using default monitoring.

It can trigger when:

- A specific event appears in the Event Log
- A performance value exceeds a threshold
- A service stops
- A process starts or stops
- A log file contains a specific entry

When the condition happens, SCOM automatically creates an alert in the monitoring console.

2. Why Create Custom Alerts

Custom alerts are useful when:

- Default monitoring does not cover a specific scenario
- You want alerts for application-specific events
- You want alerts for custom scripts or logs
- You need organization-specific monitoring

Example:

- Alert when disk usage exceeds 90%
- Alert when a specific application error appears in the Event Viewer

Steps to Create a Custom Alert

- Open the Operations Manager Console.
- Click Authoring in the navigation pane.
- Select Rules.
- Click Create a new rule.
- Choose Alert Generating Event Rule under Event Based.
- Select the target (e.g., Windows Computer) and click Next.
- Enter the rule name and choose a Management Pack.
- Configure the Event Log details (log name, event ID, source, level).
- Set the Alert Name, Severity, and Priority.
- Click Create and then Close to finish.

Troubleshooting SCOM Server

1. Check Server Health

What to check:

- CPU usage (should not be constantly >80%)
- Memory (low free RAM causes console slowness)
- Disk space (especially on C: and SQL drives)

Why it matters:

SCOM relies heavily on performance. If the server is overloaded, monitoring delays or fails.

How to check:

- Task Manager / Resource Monitor
- PerfMon counters

Fix:

- Add RAM/CPU
- Clean up disk space
- Move database/logs to separate drives

2. Verify SCOM Services

Critical services:

- System Center Management
- System Center Data Access Service (SDK)
- System Center Configuration

Why it matters:

If these stop, SCOM stops functioning.

How to check:

- Run services.msc
- Check status = Running

Fix:

- Start/restart services
- If failing check dependencies and Event Viewer

3. Check Event Viewer

Where:

- Application logs
- System logs

Look for:

- Errors from HealthService, OpsMgr SDK Service

- Database connection errors
- Authentication failures

Why it matters:

Event Viewer gives exact error messages.

Fix:

- Resolve based on error code (e.g., SQL connectivity, permissions)

4. Check SCOM Console

What to review:

- Alerts
- Active incidents
- Health state (Red/Yellow)

Why it matters:

Console shows real-time monitoring issues.

Common problems:

- Grey agents (not reporting)
- Alerts flood (misconfigured rules)

Fix:

- Drill into alert details
- Override or tune rules

5. Check Log Files

Location:

- C:\Program Files\Microsoft System Center\Operations Manager\Server\Logs

Important logs:

- OpsMgr SDK logs
- HealthService logs

Why it matters:

Logs give detailed internal errors not shown in UI.

Fix:

- Search for keywords like Error, Failed, Exception

6. Verify Database Connectivity

SCOM depends on Microsoft SQL Server

Databases:

- OperationsManager (main DB)
- OperationsManagerDW (data warehouse)

What to check:

- SQL services running
- Network connectivity
- SQL login permissions

How:

- Connect via SQL Server Management Studio (SSMS)

Fix:

- Restart SQL services
- Fix login/account permissions
- Check firewall

7. Check Management Packs

What they are:

Rules and monitors for apps (e.g., Windows, SQL, IIS)

Problems caused by:

- Outdated packs
- Corrupt packs
- Too many alerts

Fix:

- Update to latest version
- Remove unused packs
- Tune thresholds

8. Check Agent Communication

What to verify:

- Agents sending heartbeats
- No communication failures

How:

- Monitoring Agent Status

Common issues:

- Firewall blocking
- DNS issues
- Authentication failure

Fix:

- Reconnect agent
- Repair trust relationship

9. Restart Services

Why it helps:

Clears temporary glitches, memory leaks, stuck processes

Services to restart:

- System Center Management
- Data Access Service

Tip:

Restart in order:

1. SDK Service
2. Management Service

10. Check Updates and Patches

Why important:

- Fix bugs
- Improve performance
- Security fixes

What to update:

- SCOM cumulative updates
- Windows updates

Troubleshooting SCOM Clients (Agents)

1. Check Agent Installation

What to verify:

- Agent installed successfully
- Appears in SCOM console

Common issue:

- Agent not discovered

Fix:

- Manually install agent
- Use discovery wizard

2. Verify Agent Service

Service name:

- System Center Management (HealthService)

Why:

If stopped no monitoring data sent

Fix:

- Start/restart service
- Set startup type = Automatic

3. Check Agent Status in Console

States:

- Healthy
- Warning
- Critical
- Grey (not reporting)

Fix:

- Right-click Repair
- Check heartbeat failures

4. Check Event Viewer (Client Side)

Look for:

- Agent errors
- Connection failures
- Certificate issues

Fix:

- Match errors with server logs
- Resolve authentication/network problems

5. Verify Network Connectivity

Test:

ping servername

telnet servername 5723

Why:

Port 5723 is used by SCOM agents

Fix:

- Resolve DNS issues
- Open required ports

6. Check Agent Configuration

Path:

Control Panel Operations Manager Agent

Verify:

- Correct Management Server
- Correct port

Fix:

- Update configuration manually

7. Verify Firewall Settings

Required ports:

- TCP 5723 (agent communication)
- TCP 5724 (gateway if used)

Fix:

- Allow inbound/outbound rules

8. Check Log Files

Location:

- C:\Program Files\Microsoft Monitoring Agent\Agent\Logs

Important log:

- HealthService.log

Fix:

- Search for connection failures, authentication errors

9. Restart Agent Service

Why:

Resolves temporary communication issues

Command:

```
net stop healthservice
```

```
net start healthservice
```

10. Reinstall the Agent

When to use:

- Corrupt installation
- Persistent communication failure

Steps:

1. Uninstall agent
2. Delete leftover folders
3. Reinstall from SCOM server